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CLASS: IT
UNIT: SIT 351 Database Implementation.

Question 1

1. Identify a real-life entity about which data can be stored.
Student in a University System

2. Perform data modeling on the entity identified.

1. Student Entity

Attribute	Description
Student_ID	Unique identifier for each student
Name	Student name
Email	Student email
Phone	Contact number
DOB	Date of birth
Department_ID	Department the student belongs to

3. Normalize your data to 3NF.

Unnormalized Form:

Student_ID	Name	Email	Phone	DOB	Department_ID
S101	Simon	simon@mail.com	0712345678	2002-05-01	CS
S102	Mary	mary@mail.com	0722222222	2001-03-09	IT

Convert to 1NF

->Fix Atomicity Problem

Student Table (1NF)

Student_ID	Name	Email	Phone	DOB	Department_ID
S101	Simon	simon@mail.com	0712345678	2002-05-01	CS
S101	Simon	simon@mail.com	0799999999	2002-05-01	CS

Second Normal Form (2NF)

->solving Partial Dependency by Splitting the table.

Table 1 — Student

Student_ID (PK)	Name	Email	DOB	Department_ID
S101	Simon	simon@mail.com	2002-05-01	CS

Table 2 — Student_Phone

Student_ID (FK)	Phone
S101	0712345678
S101	0799999999

Now every attribute depends fully on the key.

Third Normal Form (3NF)

->solving transitive dependencies by Creating a separate Department table.

Student Table

Student_ID (PK)	Name	Email	DOB	Department_ID (FK)
S101	Simon	simon@mail.com	2002-05-01	CS

Student_Phone Table

Student_ID (FK)	Phone
S101	0712345678
S101	0799999999

Primary Key = (Student_ID, Phone)

Department Table

Department_ID (PK)	Department_Name
CS	Computer Science
IT	Information Technology

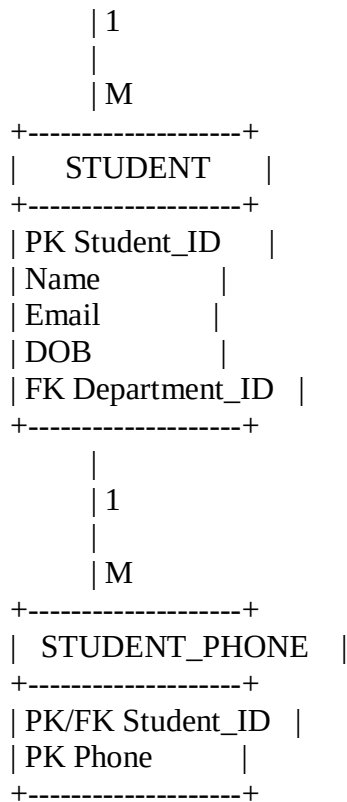
→ The Student entity was normalized to Third Normal Form by ensuring that all attributes are atomic (1NF), eliminating partial dependencies by separating phone numbers into a separate relation (2NF), and removing transitive dependencies by creating a separate Department relation linked through a foreign key (3NF). This reduces redundancy and improves data integrity.

4. Represent your data in an ERD.

```

+-----+
| DEPARTMENT |
+-----+
| PK Department_ID |
| Department_Name |
+-----+
|

```



->The ERD consists of three entities: Student, Department, and Student_Phone. The Student entity stores student information and contains a foreign key referencing the Department entity, establishing a one-to-many relationship between Department and Student. The Student_Phone entity records multiple phone numbers for students and establishes a one-to-many relationship between Student and Student_Phone.

5.Map your entities, attributes, and relationships appropriately.

DEPARTMENT

```
(
  Department_ID  PRIMARY KEY,
  Department_Name
)
```

STUDENT

```
(
  Student_ID    PRIMARY KEY,
  Name,
  Email,
  DOB,
  Department_ID FOREIGN KEY REFERENCES DEPARTMENT(Department_ID)
)
```

STUDENT_PHONE

```
(
```

```
Student_ID    FOREIGN KEY REFERENCES STUDENT(Student_ID),  
Phone,  
PRIMARY KEY (Student_ID, Phone)  
)
```

→ The ERD was mapped into relational tables by converting entities into relations and attributes into columns. Primary keys were assigned to uniquely identify each record. The one-to-many relationship between Department and Student was implemented by placing the Department_ID as a foreign key in the Student table. The multivalued attribute Phone was implemented using a separate Student_Phone relation with a composite primary key (Student_ID, Phone). This mapping ensures referential integrity and eliminates redundancy.

Question 2.

1. What type of database would you recommend for this hospital and why?

->I would recommend a Relational Database Management System (RDBMS).

Why:

Because Hospital data is highly structured thus Relational databases organize data into tables with relationships.

Because Hospitals require high reliability and RDBMS supports ACID properties.

Because Hospitals require complex queries SQL allows powerful querying and analytics.

2. How would you handle concurrent transactions to avoid deadlocks and ensure consistency?

1. Locking Mechanisms:- Locks prevent multiple users from modifying the same data simultaneously.

2. Two-Phase Locking (2PL):- This prevents inconsistent updates.

3. Deadlock Detection:- timeout, deadlock detection algorithms, and rollback one transaction.

4. Transaction Isolation Levels:- Hospitals often use Serializable or Repeatable Read for high consistency.

3. What measures should be taken to prevent data breaches and unauthorized access?

1. Role-Based Access Control (RBAC):- Different users get different permissions.

2. Encryption:- Data at Rest this Encrypt stored data in the database. Data in Transit this Use TLS/SSL encryption for network communication.

3. Authentication:- Using Strong authentication methods like:strong passwords, multi-factor authentication, and biometric login.

4. Audit Logs:- Every action should be recorded: who accessed data ? when? what changes were made? This helps detect suspicious activity.

5. Firewall and Network Security:- Protect database servers using: firewall, intrusion detection systems, and VPNs.

4. How can database backup and replication improve the performance of this system?

Database Backup:- Backup creates copies of the database this helps in Data recovery after system failure, Protection from cyber attacks, and Disaster recovery.

Database Replication:- Replication means copying data to multiple servers. This helps in:

Faster query performance:- Users access the nearest server.

High availability:- If one server fails, another takes over.

Load balancing:- Traffic distributed across servers.